International Islamic University Chittagong

Department of Computer Science & Engineering

B. Sc. in CSE Semester Final Examination, Spring 2025

Course Code: CSE-3529, Course Title: System Analysis and Design

Total marks: 50 Time: 2.5 hours

[Answer the following questions, use separate answer script for Group-A and Group-B.]

GROUP A

			Marks	CO
1.	a)	A university library management system is being developed to handle the borrowing and returning of books. Students can search for available books, borrow them using their student ID, and return them before the due date. The librarian manages the book inventory and student records. When a student borrows a book, the system records the transaction and updates the availability status of the book. If a book is returned late, a fine is calculated by the system. The system also generates monthly reports for the librarian. i) Draw the Level 0 DFD (Context Diagram) for the system. ii) Draw the Level 1 DFD, breaking down the context diagram into at least three sub-processes, such as: • Book Search and Borrowing • Book Return and Fine Calculation • Report Generation	5	CO3
	b)	An online shopping platform wants to determine whether a customer is eligible for free shipping, standard shipping, or no shipping based on the following rules: • If the customer is a premium member, they always get free shipping, regardless of the order amount. • If the customer is not a premium member, then: o If the order amount is 500 BDT or more, they get standard shipping for free. o If the order amount is below 500 BDT, then: ▪ If the customer is located within the city, they can choose standard shipping with a fee. ▪ If the customer is located outside the city, no shipping is available. Draw the Decision Tree and Decision Table for the shipping eligibility scenario.	5	CO3
		OR		
	a)	"EduPortal" is one of the most widely used online learning platforms in the country. This system bears the following characteristics: i. Students can create a personal profile with required credentials such as name, email, and password to access the platform. ii. Students can browse available courses, enroll in selected courses, and track their progress through their dashboards. iii. Instructors can upload course materials, schedule assignments,	5	CO3

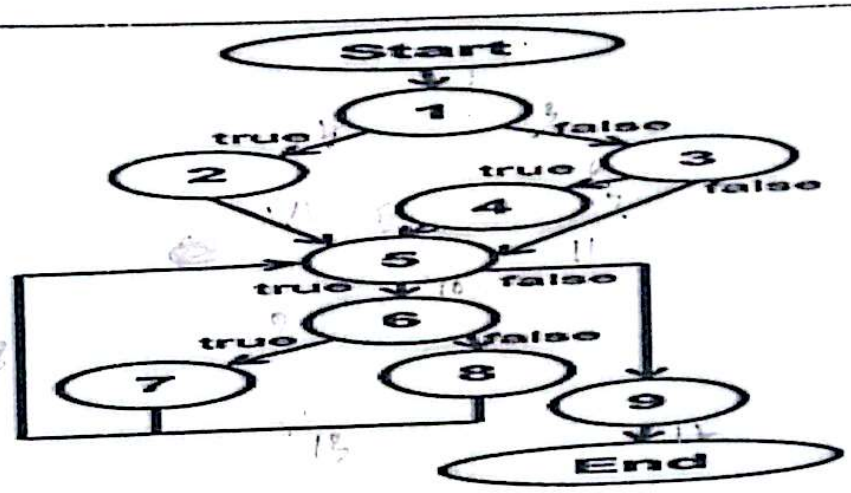
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		and view student submissions. iv. Both students and instructors must authenticate themselves before accessing any services. Based on the above scenario, let's identify: a) Find all the external entities b) Design Context Diagram or Level-0 Diagram c) Design Level-1 Diagram using balancing and leveling techniques		
	b)	**A library automation system** determines whether a member can borrow a book based on several conditions. The system evaluates the member's status and the book's availability before making a decision. 1. If **Membership Type is Student, Book is Available, No overdue,** and **Limit not reached** → *Allow Borrowing* 2. If **Membership Type is Guest,** regardless of book status → *Deny Borrowing* 3. If **Book is Reserved** or **Checked Out** → *Put on Hold* (unless Guest) 4. If **Member has Overdue Books** → *Deny Borrowing* 5. If **Borrowing Limit is Reached** → *Deny Borrowing* Based on the above scenario a. Draw a decision tree based on this scenario showing how the system reaches a decision. b. Create a **decision table** listing all combinations of conditions and the resulting decision.	5	CO3

2.	a)	Compare and contrast between top-down and bottom-up system design methodologies with proper examples and appropriate.	5	CO2
	b)	A university wants to design a database for its Online Course Enrollment System. The system must manage information about students, instructors, courses, departments, and enrollments. The system should support the following requirements: a. Students can register and enroll in multiple courses. Each student has a unique ID, name, email, and enrollment date. b. Each course has a unique course code, title, credit hours, and belongs to a specific department. c. Instructors teach one or more courses and are assigned a unique instructor ID, name, email, and designation. d. Each department offers multiple courses and is identified by a unique department ID and name. e. The enrollment of a student in a course records the date of enrollment and the grade received Based on the above scenario i. Identify all entities and their attributes. ii. Determine the relationships between entities and their cardinality. iii. Draw an Entity Relationship Diagram (ERD) representing the scenario.	5	CO4

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GROUP B

			Marks	CO
3.	a)	 1. Define cyclomatic complexity and explain its significance in software testing. 2. Calculate the cyclomatic complexity of the above control flow graph.	5	CO3
	b)	Compare and contrast Black Box Testing and White Box Testing in software engineering. a) Define both Black Box Testing and White Box Testing. b) List three key differences between them in terms of purpose, knowledge required, and techniques used. c) For the following scenarios, state whether Black Box Testing or White Box Testing is more appropriate. Justify your answer: i) Verifying the output of a login system without looking at the internal code ii) Testing all possible paths in a sorting algorithm's implementation	5	CO2
4.	a)	Explain the process of software system implementation in the software development life cycle. What are the key activities involved in the implementation phases .	4	CO2
	b)	A city hospital uses a Hospital Management System .(HMS) that handles patient records, appointments, billing, and staff scheduling. Over time, the system requires regular maintenance to ensure smooth operation and data security. The hospital's IT department is responsible for maintaining the system.The following maintenance activities are required: Fixing software bugs, corrupted data, or unexpected system crashes.Updating the system to work with new operating systems, hardware, or medical devices. Enhancing performance, adding user-friendly features, or improving the interface.Regular backups, security updates, and optimization to prevent future issues. a) Identify types of maintenance activities required for the above Hospital Management System and describe each types with respect to above scenario. b) Mention the factors that influence system maintainability	6	CO3

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5.	a)	A customer uses an online banking system to manage her accounts, transfer funds, and pay bills. The bank ensures that only authorized users can access their accounts by enforcing strong password policies and two-factor authentication. One day, the customer notices that a recent transaction shows a wrong amount — she transferred 5,000 BDT, but the system recorded 50,000 BDT. She reports this issue to the bank immediately. The bank's technical team investigates and finds that a data corruption occurred during a system update. Later that day, several users report that they are unable to access their accounts for several hours due to a denial-of-service attack on the bank's servers. • Identify how the scenario reflects the three core security goals. Explain each security goal with reference to the scenario. • Suggest one possible solution or prevention method for each of the three goals to avoid such issues in the future.	5	CO2
	b)	What is security threat? Demonstrated different types of security threats with proper example.	5	CO1
		OR		
	a)	What are the *security issues* in the context of information systems? Identify and explain **four common security issues in a computer system**	5	CO2
	b)	Explain the concept of system vulnerabilities in computing. Discuss how vulnerabilities can affect each of the following system resources with one example for each: Hardware, Software, Data.	5	CO1

Exposure, Vonnarabi lity, Threat, Controls

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Bismillahir Rahmanir Rahim

International Islamic University Chittagong

Department of Computer Science & Engineering

B. Sc. in CSE Semester Final Examination, Autumn-2024

Course Code: CSE-3529 **Course Title: System Analysis and Design**

Total marks: *50* Time: *2* hours *30* minutes

Course Outcomes (COs) of the Questions	
CO1	**Understand** project management and system development life cycle.
CO2	**Analyze** the requirement and feasibility of software application
CO3	**Design** an information system.
CO4	**Compile** the system design document

Bloom's Levels of the Questions						
Letter Symbols	**R**	**U**	**App**	**An**	**E**	**C**
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Group-A

		CO	DL
"A structured repository contains the descriptions of data flow diagram of data elements" – Identify which type of **structured analysis tools** is mentioned in the above statement. Briefly discuss about it with its significance.	2	CO2	U
Differentiate between **"Decision Tree"** and **"Decision Table"** with examples	3	CO2	U
"ShareTrip" is a trusted travel booking app in Bangladesh that provides comprehensive travel-related services. The system operates with the following features: i. Customers can create accounts by providing their personal details and preferences. Once registered, they can search for travel options, make bookings, and track their itineraries. ii. Travel Agents can register to list and manage their services, such as customized travel packages and exclusive deals. They can also interact with customers to provide tailored solutions. iii. Service Providers (e.g., airlines, hotels) manage bookings, update availability in real time, and provide updates about changes or cancellations. iv. The Admin oversees the entire system, managing accounts (customers, travel agents, and service providers), monitoring bookings, and resolving issues or disputes.	5	CO3	An

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Draw a DFD (Data Flow Diagram) according to the above scenario including all necessary symbols of DFD and the following –

a. Find all external entities

b. Design Context Diagram or Level-0 Diagram

c. Design Level-1 Diagram using balancing and leveling techniques

2.a)	State the objectives of input design. Explain the differences between good form design and bad form design with 3 different design examples.	4	CO1	U
b)	"BDJobs" is one of the largest job sites popular among different professionals as it regularly updates job information (more than 3000 valid job news are placed at any point in time) and provides a facility for job seekers to post resumes and apply online. This job portal allows users to stay updated with job market news, create personal account, sign in to their account, filter job searches by keywords, gender, and location and so on. Draw an ERD for the above scenario with proper symbols.	4	CO3	Ap p
c)	Describe the steps or phases to system design.	2	CO1	U

OR

2.a)	Design an **Entity-Relationship Diagram (ERD)** for an **Enterprise Resource Planning (ERP) system** for a **banking system**. The ERP system will help the bank manage its operations such as customer accounts, employee management, transactions, and loans.	5		

Consider the following:

- **Customer:**
  - A customer has a **customer ID, name, address, email, phone number**, and **date of birth.**
  - A customer can hold **multiple accounts** (e.g., savings, checking) and apply for **loans.**
- **Account:**
  - Each account has an **account ID, account type** (e.g., savings, checking), **balance, interest rate**, and **date opened.**
  - An account is linked to one **customer** (a customer can have many accounts).
  - Accounts can have multiple **transactions.**
- **Employee:**

An employee has an **employee ID, name, job title** (e.g., teller, loan officer, branch manager), **salary**, and **hire date**

Employees manage **transactions, customer accounts**, and **loans**

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- **Transaction:**
  - A transaction has a **transaction ID, transaction date, transaction type** (e.g., deposit, withdrawal, transfer), and **amount.**
  - A transaction is linked to an **account** and may involve an **employee** who processes the transaction.

- **Loan:**
  - A loan has a **loan ID, loan amount, interest rate, start date, end date,** and **status** (e.g., approved, pending, paid off).
  - A loan is linked to a **customer** and an **employee** (loan officer).
- **Branch:**
  - A branch has a **branch ID, branch name, branch location,** and **branch manager.**
  - Employees are assigned to a specific **branch.**
- **Payment:**
  - A payment has a **payment ID, payment date, amount,** and **payment method** (e.g., check, electronic transfer).
  - Payments are associated with **loans** (for loan repayment).

Task: Draw an **Entity-Relationship Diagram (ERD)** based on the above requirements. The diagram should include:

- **Entities** and their **attributes** (e.g., Customer, Account, Employee, Loan, etc.).
- **Relationships** between entities (e.g., a customer can have multiple accounts, an employee process transactions).
- **Cardinality constraints** (e.g., one-to-many, many-to-many) where appropriate.

Indicate **primary keys (PK)** and **foreign keys (FK)** in your diagram.

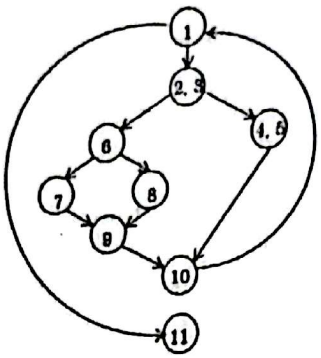
b) What is the significance of design process? Describe the necessary step of design process 3

c) Distinguish between two design methodologies like- top down strategy and bottom up strategy . 2

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Group-B

3.a) Consider the following graph, answer the following questions- 1 + 1 + 2 CO2



Flowgraph F

i. Define cyclomatic complexity.

ii. Identify node, edge and region from the above graph.

iii. Calculate cyclomatic complexity for the above flow graph in all the three ways. Also show that both ways generate same result.

b) You work as a software tester for "ABC" company that develops a new mobile messaging app. The development team has just completed the initial version of the app. Now as a software developer you are asked to conduct a test by your development team before releasing the product in Google Play Store. Your company wants to also ensure customer satisfaction, response to the new mobile messaging app as well as uncover usability issues, and assess how the app performs in diverse environments. So, for this purpose you have designed another kind of testing technique. 2 + 2 CO2

In the both above cases, as a software developer what do you think which two kinds of testing are required? Justify with proper explanation.

c) Explain the terms – **"Stress Testing"** and **"Equivalence Partitioning"** 2 CO1

4.a) List the seven common training methods for system user 2 CO1

b) Define *support* activity in implementation phase of system development process. Narrate some properties of one popular support system like EPSS. 4 CO2

c) *'Cellfin'* is an app that provides 24/7 banking services and support to the customers by providing online financial transactions. Now as a system analyst what do you think which kind of special consideration should be kept in mind for maintenance of such app? 4 CO2

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5 a)	Software and hardware selection for any organization must be done considering security and sustainability issues. As a system analyst, list the criterion for selecting hardware and software for your organization.	2	CO2	R
b)	You are given the following access control matrix. Show the Capabilities and Access Control List (ACL) from this.	3	CO2	U

	File 1	File 2	File 3	File 4
Wilson	{read, write}	{read}	--	{execute}
Taqdeer	--	{execute}	{read, write}	{read, write, execute}
Roshni	{read}	--	{write}	{execute}

c)	Write **short notes** on– i. Encryption ii. Interruption iii. Data Vulnerability iv. System Testing v. Recovery Testing	5	CO1	U

OR

5.a)	Why does system become vulnerable? Discuss different types of vulnerabilities of information resources.	2		
b)	What is control? Explain the different measures for controlling the information resources?	3		
c)	What is security? Demonstrated the different type of security issues with proper example.	5		

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International Islamic University Chittagong
Department of Computer Science and Engineering
B.Sc. in CSE Final Examination, Spring-2024
Course Code: CSE-3529 Course Title: System Analysis and Design
Marks: 50 Time: 2 Hours 30 Minutes

Instructions: *Answer all the questions. Figures on the right margin indicate marks.*

GROUP: A

1	a)	Suppose, "MetroTrain" is one of the most popular metro train systems in the world. This system bears the following characteristics: i. Users can register for a personal account with the necessary personal information, username, and password to access the metro train services. ii. Users can purchase tickets, check train schedules, and view other related information from their personal profiles. iii. All users have to log in to the system going through an authentication process. Based on the above scenario, let's identify: A. Find all the external entities B. Design Context diagram or Level-0 Diagram C. Design Level-1 diagram using balancing and leveling techniques	5	CLO3
	b)	An IT company uses a project management system to assign projects to teams based on several criteria. The system evaluates project requirements and team capabilities to make decisions. The criteria used for decision-making include: • Project Complexity (Complex, Moderate, Simple) • Team Expertise (High, Medium, Low) • Deadline Urgency (Urgent, Standard, Flexible) • Budget Availability (High, Medium, Low) Based on these criteria, the system assigns the project to one of three teams: **Team A:** Suitable for complex projects with high expertise and urgent deadlines. **Team B:** Suitable for moderate projects with medium expertise and standard deadlines. **Team C:** Suitable for simple projects with low expertise and flexible deadlines. Your task is to draw a decision tree and a decision table based on this scenario.	5	CLO3

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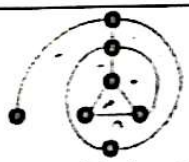
2	a)	Distinguish between bottom-up and top-down system design methodologies with proper examples and figures.	5	CO2
	b)	An airline reservation system is designed to manage flights, passengers, bookings, and other related information. The system handles various aspects of airline operations, including flight schedules, passenger details, and seat reservations. The key elements of the system include: **Airlines:** Each airline operates multiple flights. **Flights:** Each flight has a unique flight number, departure and arrival airports, and a scheduled departure and arrival time. **Passengers:** Passengers book flights and each passenger has a unique passenger ID, name, contact information, and passport number. **Bookings:** Bookings are made by passengers for flights. Each booking has a unique booking ID, booking date, and status (confirmed, canceled, pending). **Seats:** Each flight has a specific number of seats, each with a seat number and class (economy, business, first class). Based on this scenario, you need to draw an Entity-Relationship Diagram (ERD) with the proper symbol.	5	CO4
		OR		
	a)	There are three types of accounts in a bank. Assume that the accounts are current account, saving account and fixed deposit account. In current account bank do not give any interest, in saving account bank gives 5% of interest, and in fixed deposit account bank gives interest depend on time at following: **Duration** **Interest** **6 months** **8%** **1 year** **10%** **5 years** **12%** Draw the decision table, decision tree.	5	CLO3
	b)	A company wants to develop a ride-hailing system similar to Uber. The system needs to manage drivers, customers, rides, and vehicles. The following details are provided for the system: Drivers: Each driver has a unique driver ID, name, email, phone number, and license number. Customers: Each customer has a unique customer ID, name, email, and phone number. Vehicles: Each vehicle has a unique vehicle ID, license plate number, make, model, and year. Rides: Each ride has a unique ride ID, a pickup location, a drop-off location, a fare amount, and a timestamp for the start and end of the ride.	5	CLO3

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		Each ride is associated with one driver, one customer, and one vehicle. Based on the provided scenario, develop an ERD that captures the relationships between drivers, customers, rides, and vehicles. The diagram should include entities, attributes, primary keys, foreign keys, and relationships with appropriate cardinalities.		

GROUP: B

3	a)	 a) Considering the above graph, answer the following question; i. Define flow graph. ii. Identifying the node, edge and region from the above flow graph. iii. Calculate the cyclomatic complexity from the above flow graph at least two ways and also show that both ways generate the same result.	5	CLO2
	b)	b) What is white box testing and black box testing? Describe with an example.	5	CLO3

4	a)	a) What is system maintenance? Demonstrate the four types of system maintenance with necessary examples.	5	CLO2
	b)	b) An EdTech company has developed a Learning Management System (LMS) to facilitate online education for schools and universities. The system includes functionalities for course management, user management (students, teachers, and administrators), content delivery, assessments, and reporting. Over time, as the system is used, it requires maintenance to fix bugs, add new features, and improve performance. write the system maintenance steps for the above scenario	5	CLO3
		OR		
	a)	Explain the process of software system implementation in the software development life cycle. What are the key activities involved in the implementation phase?	3	CLO2
	b)	What are the different types of software maintenance activities? Provide examples of corrective, adaptive, perfective, and preventive maintenance.	3	CLO2
	c)	Discuss the significance of testing in software system implementation. What types of testing should be conducted during the implementation phase, and how do they contribute to the overall quality of the software system?	4	CLO2

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5	a)	Write some criteria for software selection and hardware selection	2	CLO1
	b)	Outline the difference between a security vulnerability and a security threat. Provide examples of common security vulnerabilities and threats faced by organizations today.	4	CLO4
	c)	Write short notes on: (Any two) i) Alpha Beta testing ii) Interruption, Interception iii) Computer Criminals iv) Security goals	4	CLO1

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International Islamic University Chittagong
Department of Computer Science and Engineering
B.Sc. in CSE Final Examination, Spring-2023
Course Code: CSE-3529 Course Title: System Analysis and Design
Marks: 50 Time: 2 Hours 30 Minutes

Instructions: *Answer all the questions. Use separate answer script for Group A and Group B. Figures on the right margin indicate marks.*

Group: A

1(a)	"A structured repository contains the descriptions of data flow diagram of data elements" – Identify which type of structured analysis tools is mentioned in the above statement. Briefly discuss about it with its significance.	2	CO3
1(b)	Differentiate between "Decision Tree" and "Decision Table" with examples.	3	CO3
1(c)	We wish to draw the DFD for a distance education university. The enrolment process works as follows: Students send in an application form containing their personal details and their desired course. The university checks that the course is available and the student has necessary academic qualifications. If the course is available the student is enrolled in the course and the university confirmed the enrolment by sending a confirmation letter to the student. If the course is unavailable the student is sent a rejection a letter. (i) Find out and describe the suitable DFD context level input and output. (ii) Find out the level 0 DFD	5	CO3

OR

"Amazon.com" is a globally reputed e-commerce platform that offers an extensive and diverse range of products such as clothing, electronic gadgets, household items and many more. This site has the following criterion:

i. Users/Customers can create accounts, providing personal information, usernames, and passwords for secure access.

ii. Sellers needs to list products, including associated media (images, descriptions), for customers to view and purchase.

iii. Registered users, both customers and sellers, interact with the Amazon online shopping system after authentication, including browsing and shopping.

Draw a DFD (Data Flow Diagram) according to the above scenario including all necessary symbols of DFD and the following –

A. Find all the external entities
B. Design Context diagram or Level-0 Diagram
C. Design Level-1 diagram using balancing and leveling techniques

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2(a) State the objectives of input design. Explain the differences between good form design and bad form design with 3 different design examples. 5 CO3

2(b) A department of a company has a supervisor and at least one employee. Employees must be assigned to at least one, but possibly more departments. At least one employee is assigned to a project, but an employee may be on vacation and not assigned to any projects. The important data fields are the names of the departments, the projects, the supervisors and the employees, as well as the supervisor and employee number and a unique project number. Draw the ER diagram related to the scenario described above and provide short explanation of the diagram. 5 CO3

OR

"IIUC Computer Club" is one of the most active and promising student clubs of Department of Computer Science & Engineering at International Islamic University Chittagong. This CSE club dedicatedly works for students and organizes different events that enhances the academic curriculum. Students of CSE department by default gets its membership and participates in various activities organized by club.
Draw an **ERD** for the above scenario with proper symbols.

Group: B

3(a) What is system testing? Is testing compulsory for a system or not? Justify your answer with example. 5 CO3

OR

Assume you are a software engineer of a developing firm. Write the possible consequences may occur if you do not make any kind of testing before handover your software (Discuss for both user and developer point of view).

3(b) Define the following terms with suitable example: Alpha testing, Beta testing, White box testing and Black box testing. 5 CO3

4(a) State system implementation. Mention the six major activities that ensures system implementation process. 2 CO1

4(b) Write about the types of documentation of any system with example. 4 CO1

4(c) Mr. XYZ is a project manager of a mobile game's development software firm. He has a dedicated group of teams that works for app maintenance. Now if any maintenance request comes to his organization, what do you think how will the project manager handle it? Explain with figure. 4 CO2

5(a) What is threat? Briefly explain about different kinds of threats. Write some criteria for software and hardware selection. 5 CO2

5(b) What is security? Write some security issues with examples. How to prevent vulnerability from a system? 5 CO2

OR

Write short notes on– i) Encryption ii) Interruption iii) Data Vulnerability iv) System Testing iv) Recovery Testing

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International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Final term Examination, Spring 2023

Course Code: CSE 3529 Course Title: System Analysis and Design

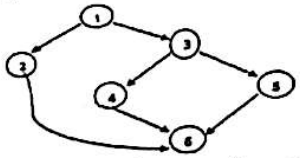
Time: 2-hours 30 minutes Total marks: 50

[Answer all the questions. Figures in the right-hand margin indicate full marks.]

Group A

		M	CO
1.a)	Suppose, "X" is one of the most popular social networking sites in the world. This site bears the following characteristics: i. Users can open their personal account with necessary personal information, username and password to communicate with other registered users in the social media platform ii. Users can upload photos, videos and other types of media from their own personal profile iii. All users have to login in the site going through authentication process Draw a DFD (Data Flow Diagram) according to the above scenario including all necessary symbols of DFD and the following – A. Find all the external entities B. Design Context diagram or Level-0 Diagram C. Design Level-1 diagram using balancing and leveling techniques	5	CO3
b)	Assume you own a software firm named "T". To describe the workflow of your company a DFD has already been created. Considering the above context – What do you think which type of structured analysis technique will be suitable to keep record about the detailed descriptions of DFD? Explain briefly with example. ***Or,*** Describe the phases to design any kind of system.	2	CO3
c)	Define decision trees in structured analysis. Explain the mechanism of a decision table with proper example. ***Or,*** What are the four DFD symbols? Illustrate the rules for drawing DFD.	3	CO4
2.a)	State the objectives of a good form design. How can you differentiate between a good form design and a bad form design? Explain with an example. ***Or,*** Distinguish between bottom-up and top-down system design methodologies.	5	CO1
b)	*"Daraz"* is one of the leading e-commerce platforms in Bangladesh. Users can create their personal account in this site. It offers online marketplace for various products of different brands. With a few clicks people can select their products and pay for it easily. Draw an ERD for the above e-commerce site with proper symbols.	3	CO4
c)	Describe the steps in system's database design.	2	CO1

Group B

3.a)	 Considering the above graph, answer the following questions- i. Define flow graph. ii. Identify node, edge and region from the above graph. iii. Calculate cyclomatic complexity for the above flow graph in both ways. Also show that both ways generate same result. ***Or,*** How can calculate cyclomatic complexity using graph matrix and connection matrix? Illustrate with an example.	5	CO2

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b) The requirement specification of a web-based software application stated that maximum 100 users may access simultaneously. After deployment the application failed to response properly when 90 users accessed at a time. It was recovered later and then on a certain occasion 150 users tried to access and again it crashed. Which two software testing methods could prevent those two failures respectively?

c) Briefly explain the two types of Black Box testing techniques with example.

4.a) Outline the success factors of implementation phase. Also name the six major types of activities required to accomplish the system implementation phase. 2

b) Demonstrate the four types of system maintenance with necessary example. 4

c) *"MedEasy"* is a website that provides 24/7 customer service and support to the customers by providing online pharmaceutical products. So, from the perspective of a system analyst, what do you think which kind of special considerations should be kept in mind for maintenance of such a website? 4

5.a) There are 4 types of issues in case of ensuring security: Relate those security issues for a university website like IIUC and mention them briefly. 2

b) In order to ensure a system's smooth performance, hardware and software design is of utmost importance. Write some criteria for software selection and hardware selection. 3

c) A system can be protected or secured from all security threats using four primary methods. State the methods and explain each of them shortly. 5

Or,

Explain the following terms –

i. Confidentiality
ii. Interception
iii. Data Vulnerability
iv. Alpha testing
v. Validation testing

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International Islamic University Chittagong (IIUC)

Dept. of Computer Science & Engineering (CSE)
Final Examination: Spring-2021
Program: B.Sc. in CSE

Course Code: **CSE-3529** Course Title: **System Analysis & Design.**
Time: 2.5 Hours Full Marks: 50

All questions should be answered.

Group A

CO3 1.(a) The passenger is required to fill in a reservation form giving details of his journey. The counter clerk ensures whether the place is available. If so, entries are made in the register ,tickets are prepared ,amount is computed and cash is accepted. A booking statement is prepared in triplicate from the reservation register. One copy of it is retained as office copy ,the other is pasted on the compartment and the third is passing on to the train conductor .Besides booking statement, cash statement is prepared at the end of each shift. Prepare a DFD to describe this situation., Give the following terms for the above "Railway Reservation System": (i) Input design of the system (ii) Draw Data Flow Diagram(DFD) (iii) Draw context free diagram (iv) Output design of the system — 5

CO3 1.(b) Construct a decision tree and calculate Expected Value (EV) based on the values in the table below: Formula: $EV=\sum P(Xi)\times Xi$ — 5

Alternatives	Growing	Declining
Stocks	70	-13
Mutual Funds	53	-5
Bonds	20	20
Probability	0.4	0.6

Or

CO3 1.(a) When CASE keyword is used in Structure English? Describe with example. Name the popular CASE tools. State the advantages and Disadvantages of Indexed Sequential File Organization and Direct File Organization — 5

CO3 1(b) Draw a decision tree and decision table for a project management — 5

CO3 2.(a) What is input design and its objectives — 3

CO3 2.(b) What is form design? Objectives of good form design. — 4

CO3 2.(c) What is database design? What is the logical design of a database? — 3

Group B

CO3 3.(a) What is system testing? Is testing compulsory for a system or not? Justify your answer — 3

CO3 3.(b) What is white box testing and black box testing? Describe with example. — 5

CO3 3.(c) What are the difference between Alpha testing and Beta testing? — 2

CO4 4.(a) What is system maintenance? Why it is necessary? — 3

CO4 4.(b) Discuss the process of system maintenance. — 3

CO4 4.(c) What are the types of system maintenance? Distinguish between adaptive and corrective maintenance? — 4

CO2 5.(a) Write some criteria for software selection and hardware selection — 2

CO2 5.(b) What is Threats? Briefly discuss about physical threat and non-physical threat. — 4

CO2 5.(c) Write short notes on the following: i)Network security ii) End-point security iii) Internet security iv) Cloud security — 4

Or

CO2 5.(a) What is meant by "Project Management"? Why it is needed? Write short note on: (i)PERT (iii)ROI — 3

CO2 5.(b) Prepare Data Dictionary entries for i)The data elements Basic Pay ,Professional Tax ii)The data stores Salary Summery File, Arrears Files iii)The data flows Earning details, Deduction details — 4

CO2 5.(c) Write the importance of documentation in software development, — 3

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